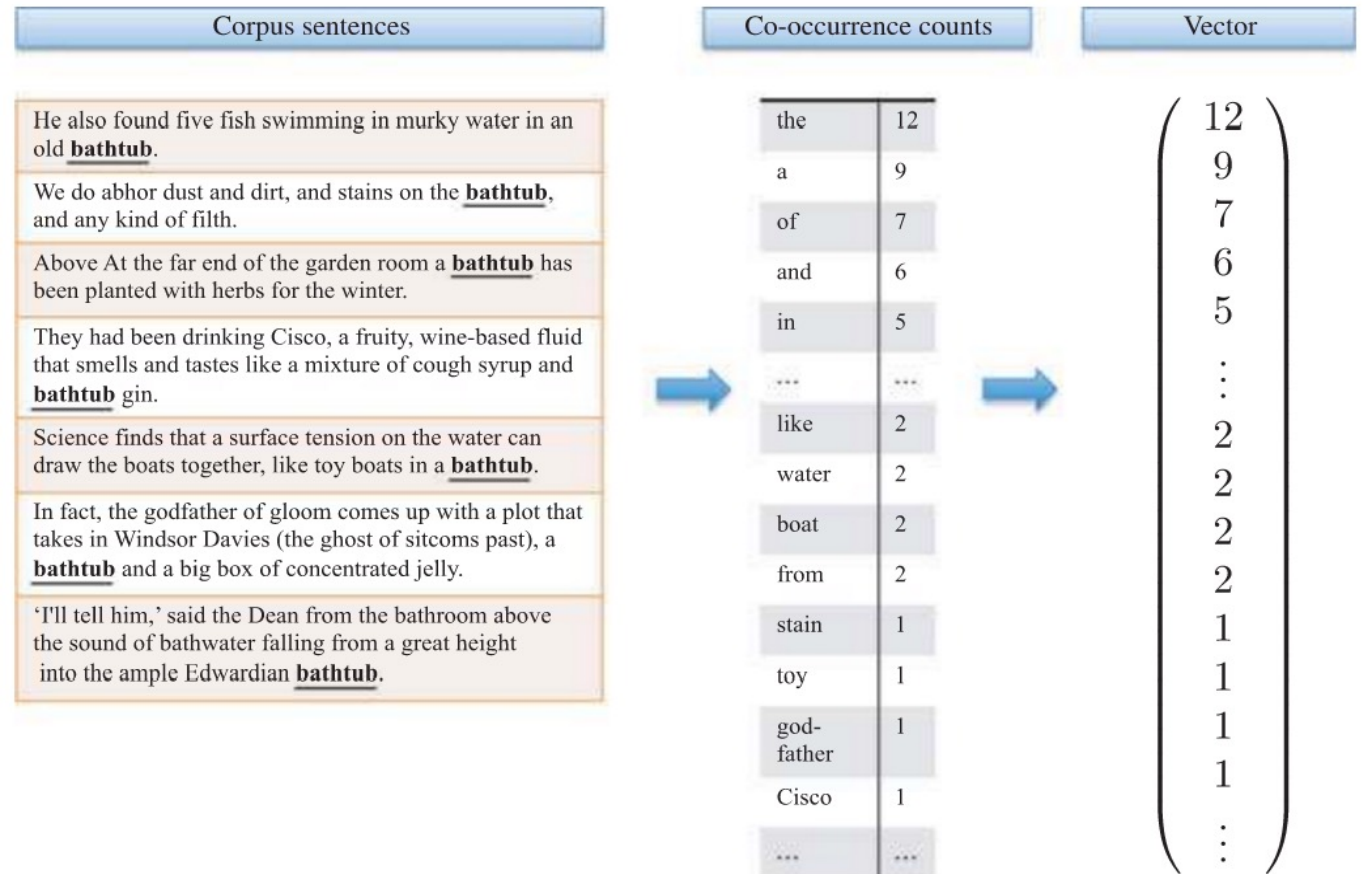


Word Embeddings: Count-based Models

- Contexts are defined as neighboring words
 - Usually in a window of $\pm K$ words
- Dimensions correspond to context wordforms
- Values in entries correspond to counts – the number of times a word and a context word co-occurred



How Do These Vectors Represent Meaning?

- Similarity can be measured using vector distance metrics
- A popular choice is the “cosine similarity”:

$$\text{similarity}(w, u) = \frac{w \cdot u}{\|w\| \|u\|} = \frac{\sum_{i=1}^n w_i u_i}{\sqrt{\sum_{i=1}^n w_i^2} \sqrt{\sum_{i=1}^n u_i^2}}$$

How Do These Vectors Represent Meaning?

- Instead of representing words with 1-hot vectors (words are either the same or unrelated), embed them in a space that reflects their similarity patterns
- But taking neighboring wordforms as features is a bit naïve...

